

Comprehensive Literature Review

Phishing detection continues to be an active area of cybersecurity research, with a range of techniques proposed, such as rule-based logic, heuristic models, blacklist systems, and machine learning.

Since our web app relies on rule-based detection and blacklist APIs, this review explores relevant studies supporting our approach.

→ Rule-Based Detection Approaches

Basnet et al. (2015) proposed a phishing detection method that uses rule-based logic to analyze website behavior and identify phishing attempts. The system examined attacker tactics and developed rules to distinguish phishing from legitimate webpages. Their approach achieved an impressive accuracy of 95–99% with a false positive rate as low as 0.5–1.5%.

This high accuracy proves the effectiveness of rule-based systems, especially when rules are carefully designed. It also validates our choice of using a rule-based engine in our mobile app to detect malicious links.

→ Heuristic and Blacklist Hybrid Schemes

Researchers at Temple University introduced “MobiFish,” a mobile anti-phishing system that relies on heuristic rules combined with blacklist checks. Their research showed that while these approaches are effective in many cases, they still struggle with detecting zero-day phishing attacks (previously unknown threats).

However, for known and pattern-based phishing links, the combination of heuristics and blacklist APIs is still one of the most reliable and lightweight solutions, making it ideal for mobile apps like ours.

→ Rule-Based Web Detection Systems

Recent implementations (referenced via Gale) designed for browser-based detection rely on lightweight rules to analyze URL structures and flag suspicious links in real-time.

These systems are easy to integrate into web applications and support low-latency responses, reinforcing the practicality of our approach

→ Comprehensive Survey of Detection Techniques

Dholakia and Agrawal (2018) reviewed many phishing detection methods and concluded that no single method solves all threats.

Blacklist and rule-based systems effectively handle known attacks, but zero-day and sophisticated phishing require machine learning or deep learning integration. This supports our plan to begin with rule-based logic and later integrate ML for enhanced detection.

→ Rule-Based SMS-Based Detection (Analogous Techniques)

Akande et al. (2021) developed a real-time smishing detector using rule-based classifiers like RIPPER and C4.5.

Although SMS-focused, the underlying method, scanning textual links for suspicious patterns, is directly applicable to web URLs, especially in resource-constrained environments.

References & Research

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